Cross-Modal Context-Aware Learning for Visual-Prompt-Guided Multimodal Image Understanding in Remote Sensing

Xu Zhang, Jiabin Fang, Zhuoming Ding, Jin Yuan, Xuan Liu, *Member, IEEE*, Qianjun Zhang, and Zhiyong Li, *Member, IEEE*

***Abstract*—Recent advances in image understanding have enabled methods that leverage large language models (LLMs) for multimodal reasoning in remote sensing (RS). However, existing approaches still struggle to steer models to the user-relevant regions when only simple, generic text prompts are available. Moreover, in large-scale aerial imagery, many objects exhibit highly similar visual appearances and carry rich interobject relationships, which further complicates accurate recognition. To address these challenges, we propose cross-modal context-aware learning for visual-prompt-guided multimodal image understanding (CLV-Net). CLV-Net lets users supply a simple visual cue—a bounding box—to indicate a region of interest, and uses that cue to guide the model to generate correlated segmentation masks and captions that faithfully reflect user intent. Central to our design is a context-aware mask decoder (CMDecoder) that models and integrates interobject relationships to strengthen target representations and improve mask quality. In addition, we introduce a semantic and relationship alignment (SRAlign) module: a cross-modal semantic consistency loss enhances fine-grained discrimination among visually similar targets, while a relationship consistency loss aligns textual relations with visual interactions, identifying meaningful commonalities to guide model output. Comprehensive experiments on two benchmark datasets show that CLV-Net outperforms existing methods and establishes new state-of-the-art results. The model effectively captures user intent and produces precise, intention-aligned multimodal outputs. The code will be publicly available at: https://github.com/XuZhang1211/CLV-Net**

***Index Terms*—Multimodal alignment, multimodal large language model (MLLM), remote sensing (RS), scene understanding.**

I. Introduction

IMAGE understanding—the extraction of essential semantic information from images via visual and textual representations—underpins a wide range of visual tasks such as object detection [1], image captioning [2], [3], and segmentation [4], [5], [6]. These technologies have found broad application in human–computer interaction [7], [8], autonomous driving [9], [10], [11], and medical diagnosis [12]. Recent advances in large language models (LLMs), and especially multimodal LLMs (MLLMs), have shifted image understanding from purely unimodal perception toward multimodal, collaborative comprehension. This paradigm shift builds upon foundational progress in broader cross-modal learning, where recent advancements in multimodal feature fusion [13], representation learning [14], [15], and synergistic knowledge transfer [16] have demonstrated significant success across various visual understanding tasks. Furthermore, advanced regularization and inference techniques have been instrumental in optimizing these complex multimodal feature spaces [17]. By exploiting cross-modal correlations and the strong reasoning capabilities of MLLMs, these approaches not only improve accuracy but also reveal deeper semantic structures in scenes, enabling richer, context-aware interpretations of visual content and delivering notable gains in natural scene understanding [18], [19], [20], [21], [22], [23].

Unlike natural images, remote sensing (RS) imagery covers vast areas and often contains targets with subtle, low-contrast features, which greatly increases the difficulty of precisely focusing on and interpreting scene content. Recent work has begun applying MLLMs to collaborative understanding in RS, including GeoPix [24], GeoChat [25], GeoGround [26], and GeoPixel [27]. For example, GeoPixel [27] uses textual prompts to guide an LLM in producing global semantic descriptions, which are then leveraged for reasoning-aware segmentation to generate both textual summaries and corresponding target masks [see Fig. 1(a)]. Although this yields useful high-level scene semantics, it lacks a fine-grained understanding of local regions. To address that gap, GeoPix [24] augments textual descriptions to support multireferring segmentation and visual question answering, enabling more detailed localization of local areas [see Fig. 1(b)]. Nevertheless, GeoPix relies on experts to scan vast imagery, identify regions of interest, and craft referring expressions, which imposes substantial cognitive and interaction burdens on users.

Prevailing methods predominantly rely on textual prompts to generate global captions and segmentation masks. While some approaches incorporate visual prompts (e.g., bounding boxes), they typically restrict their analysis to the interior

Received 5 November 2025; revised 28 January 2026 and 1 March 2026; accepted 10 March 2026. Date of publication 13 March 2026; date of current version 24 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62406263, Grant 62272157, and Grant U23A20341. *(Corresponding authors: Jin Yuan; Qianjun Zhang.)*

Xu Zhang, Jiabin Fang, Zhuoming Ding, Jin Yuan, and Xuan Liu are with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China (e-mail: yuanjin@hnu.edu.cn).

Qianjun Zhang is with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Sichuan 611756, China (e-mail: zqjblue@foxmail.com).

Zhiyong Li is with the School of Robotics and the National Engineering Research Center of Robot Visual Perception and Control Technology, Hunan University, Changsha 410082, China.

Digital Object Identifier 10.1109/TGRS.2026.3673739

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

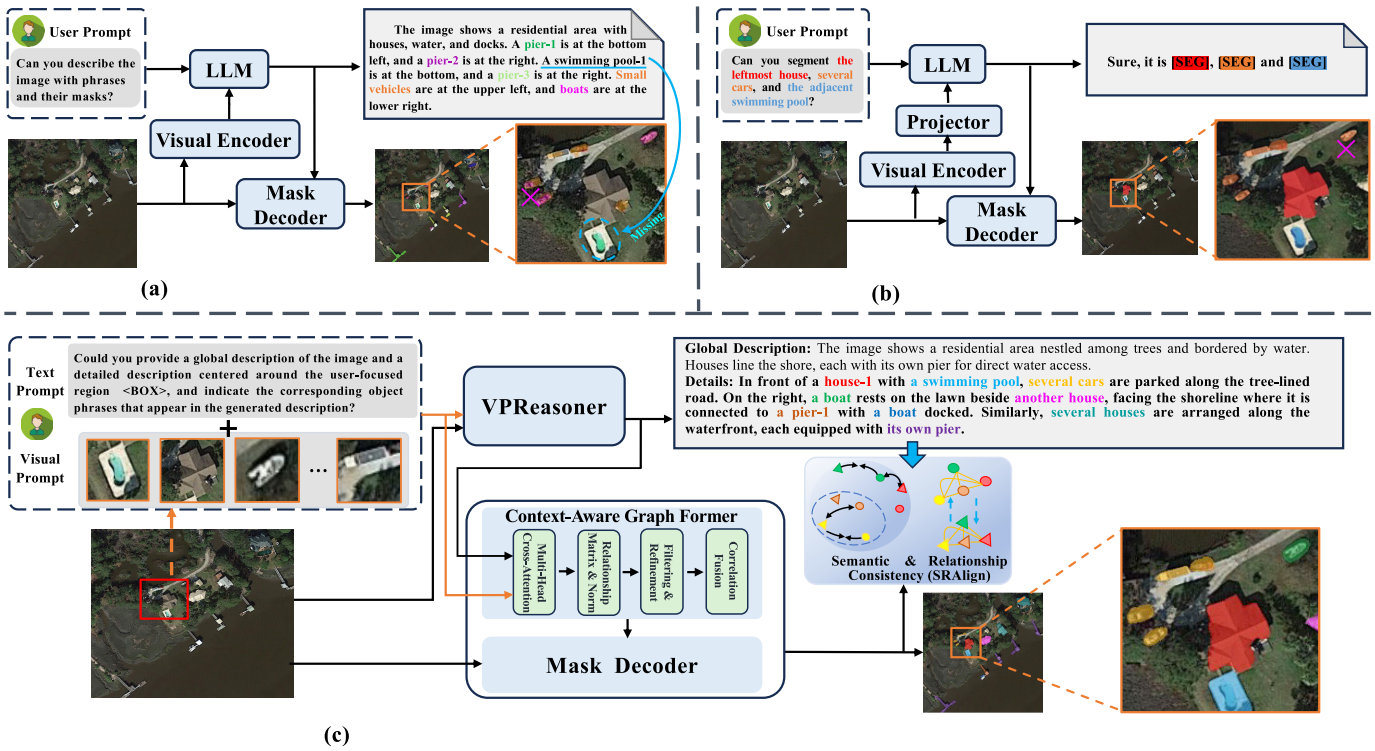


Fig. 1. Framework Comparison between our method and the existing methods. GCG (a) directly generates a global summary on an image, while multireferring segmentation (b) allows detailed VQA and segmentation on local regions conditioned on a textual prompt. Both approaches predict wrong labels for masks due to the similar visual appearances between objects. Differently, our approach (c) receives a simple visual-prompt embedded into a textual prompt to generate a concise global and detailed local descriptions along with accurate masks corresponding to caption tokens. Underlined text denotes predicted words with missing masks, while crosses indicate mismatches between the generated text and the target.

of the prompted region. This localized constraint fundamentally limits the model's receptive field, depriving it of the broader global context. In complex RS scenarios, the accurate identification and semantic understanding of objects heavily depend on surrounding environmental cues, background topologies, and interobject relationships. By discarding global semantics, models relying solely on confined prompts often struggle to resolve visual ambiguities—particularly when targets share similar textures with their surroundings, or when local visual features are degraded by occlusion and lighting variations. To overcome this problem, practical RS interpretation frequently demands a dual perspective: users need to grasp the global scene while simultaneously scrutinizing specific regions of interest. Therefore, establishing an effective mechanism to seamlessly integrate global semantic descriptions with fine-grained local reasoning is critical.

This article presents a visual-prompt-guided multimodal framework for RS image understanding. In addition to a global textual prompt, our system accepts a simple visual prompt—e.g., a bounding box—that specifies a region of interest. The model then generates a compact global semantic caption plus several detailed local descriptions, each paired with a segmentation mask [see Fig. 1(c)]. By combining visual and textual cues, the approach enables precise, user-focused interpretation while substantially reducing user effort. It also strikes a balance between comprehensive global understanding and fine-grained local analysis.

To implement this framework, we propose cross-modal context-aware learning for visual-prompt-guided multimodal image understanding (CLV-Net), which comprises two principal modules: a visual-prompt scene reasoner (VPReasoner) and a context-aware mask decoder (CMDecoder). As shown in Fig. 1(c), the VPReasoner ingests a global image together with visual and textual prompts, fuses these bimodal cues, and conditions an LLM to generate hierarchical captions—a concise global summary alongside detailed local descriptions. Building on this, the CMDecoder explicitly models cross-modal relationships between textual and visual representations and captures contextual dependencies among objects. Cross-modal modeling yields precise alignment between mask regions and caption tokens, while contextual modeling reduces object classification errors by exploiting neighboring object context. Consequently, the CMDecoder produces mask outputs whose object labels accurately correspond to the generated captions. To narrow the representation gap between caption tokens and object masks, we introduce two novel training objectives: a semantic consistency loss and a relationship consistency loss. The semantic consistency loss pulls cross-modal representations of the same object closer while pushing apart representations of different objects. The relationship consistency loss facilitates cross-modal relational alignment by leveraging valuable commonalities between the textual and visual relationships, helping mitigate category and boundary ambiguities during model output. Rather than enforcing a strict one-to-one correspondence, our approach employs supervised learning to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

refine both relationship matrices, guiding the model effectively by leveraging valuable commonalities. Together, these losses improve the model's ability to discriminate between objects and ensure consistent correspondence between caption tokens and masks. Extensive evaluations on the RS dataset GeoPixelD [27] and the natural-image dataset GranD [22] demonstrate the effectiveness of our framework, which achieves state-of-the-art performance compared with existing methods.

In summary, the main contributions of this work are as follows.

1) We propose CLV-Net. Given a simple visual prompt that specifies a local region in the image, CLV-Net enables the generation of a concise global summary together with detailed local descriptions and their corresponding masks—reducing user effort while producing richer, intention-aligned semantic outputs.
2) We design a CMDecoder that explicitly models and fuses interobject semantic relationships to enhance object representations and improve mask quality. To further bridge modalities, we introduce two novel training objectives that enforce cross-modal representation and relationship consistency, yielding precise correspondence between caption tokens and masks.
3) Extensive experiments on the GeoPixelD RS dataset and the GranD natural-image dataset demonstrate the effectiveness and generalizability of our approach, achieving new state-of-the-art results across diverse domains.

II. Related Work

A. Multimodal Large Language Models

LLMs have emerged as powerful tools for bridging vision and language, enabling joint reasoning across modalities. The effectiveness of these models is deeply rooted in sophisticated cross-modal feature alignment and representation learning strategies [13], [14], [15]. Early efforts extended language models with visual backbones to perform tasks such as captioning [28], [29], visual question answering [30], [31], [32], and dialog-based scene understanding [2], [18]. More recent research has evolved from single-modality processing toward MLLMs [33], [34], [35], which integrate visual and linguistic understanding within a unified framework to achieve fine-grained grounding and pixel-level reasoning through the reasoning capabilities of LLMs. For instance, GSVa [36] introduces generalized segmentation via MLLMs, and Groma [37] proposes localized visual tokenization for fine-grained grounding. LISA [20] is the first framework to leverage semantic text generated by LLMs for reasoning-based segmentation, inaugurating the LLM-based collaborative understanding paradigm. Building upon LISA, GLaMM [22] extends this paradigm to enable multiobject collaborative understanding scenes, producing coordinated outputs across multiple targets within an image. PixelLM and GROUNDHOG [21], [38] further advance pixel-level reasoning with LLMs, while STVGBert [39] extends visual grounding into spatiotemporal video contexts. More recently, OMG-LLaVA [40] unifies reasoning across image-level, object-level, and pixel-level granularity, demonstrating the growing versatility of MLLMs in multimodal understanding.

Despite these advances, most existing methods rely on generic textual prompts for grounding, which often lack the granularity and precision needed to capture user intent in complex visual scenes. In contrast to region-based image understanding methods [41], [42], which restrict the output strictly to the content within a bounding box, our method leverages the bounding box as an anchor to locate the region of interest while generating relation-aware descriptions that extend beyond the box. This enables the model not only to focus on the targets of interest but also to capture connections between these targets and the surrounding context.

B. RS Multimodal LLMs

RS presents unique challenges for multimodal reasoning due to its large-scale, high-resolution imagery and dense object distributions. Prior to the widespread adoption of LLMs, specialized networks achieved notable progress in addressing these challenges. For instance, HCNet [43] and S^2CPNet [19] introduced hierarchical aggregation and semantic-spatial collaborative perception to enhance cross-modal alignment for image captioning. Similarly, C^2Net [44] demonstrated the value of global–local context perception and cross-scale interaction for precise segmentation. Bridging specialized architectures with foundation models, PE-RSIC [45] further explored parameter-efficient transfer learning to adapt vision–language pretraining to the RS domain.

However, despite their effectiveness, specialized models often lack the extensive prior knowledge and robust reasoning capabilities inherent in LLMs. Meanwhile, the substantial domain gap between natural and RS images limits the direct applicability of general-purpose models. To bridge this gap, recent efforts have extended MLLMs to the RS domain. RemoteCLIP [46] pioneers a vision–language foundation model for RS, learning semantically rich and transferable representations through large-scale pretraining on diverse RS datasets. LRS-VQA [47] introduces text-guided token pruning with a dynamic image pyramid for efficient processing of large RS imagery and establishes a benchmark for RS question answering. EarthGPT [29] integrates multisensor data, including optical, SAR, and infrared, through visual-enhanced perception and mutual comprehension mechanisms, supported by the MMRS-1M dataset for instruction-based learning. RSGPT [48] emphasizes large-scale data curation and constructs the RSICap dataset to enhance vision–language alignment for RS captioning. Furthermore, SkyEyeGPT [49] unifies multiple RS vision–language tasks with instruction-tuning across single- and multi-task conversations to improve adaptability. SkySenseGPT [50] advances fine-grained reasoning through the FIT-RS dataset, achieving state-of-the-art performance across RS benchmarks. SkySense V2 [51] further refines this paradigm with a unified Transformer backbone and self-supervised pretraining, leveraging adaptive patch merging and learnable modality prompts for efficient multimodal learning. Graph-based models [52], [53] such as STAR enhance contextual reasoning by modeling spatial and semantic relations among RS objects. Large-scale datasets

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

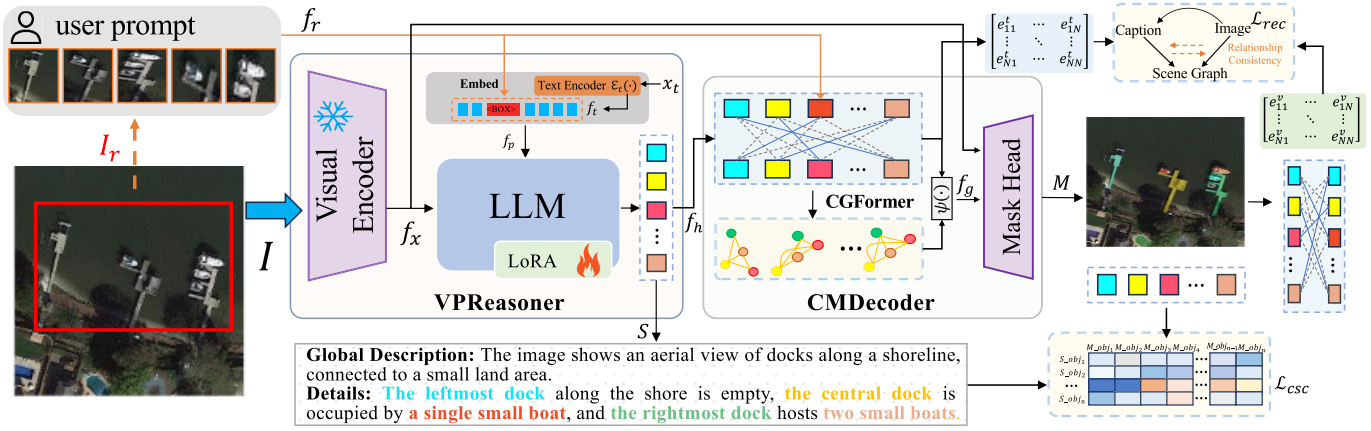


Fig. 2. Overview of the proposed CLV-Net, comprising three components: a VPReasoner that uses a bounding box to guide content generation for the user-specified region; a CMDecoder that incorporates interobject relationships to enhance target representations and mask quality; and an SRAlign module that enforces fine-grained discrimination via cross-modal semantic consistency loss and maintains visual–textual relationship consistency via relationship consistency loss.

like RS5M and GeoRSCLIP [54] further support general-purpose RS multimodal training and evaluation. Collectively, these developments establish a comprehensive foundation for vision–language understanding in RS and drive the evolution of RS MLLMs.

C. Visual Grounding LLMs in RS

Relying solely on textual descriptions from LLMs fails to provide users with rich and detailed explanations of image content. To enhance users' understanding of RS imagery, recent works have extended RS LLMs to include pixel-level visual grounding and segmentation, providing text descriptions alongside corresponding target regions for improved accuracy. GeoChat [25] is the first versatile RS vision–language model (VLM) offering multitask conversational capabilities with high-resolution RS images. It can answer image-level queries and engage in region-specific dialogs, grounding objects by referencing their spatial coordinates. GeoGround [26] introduces a unified framework for horizontal bounding box (HBB), oriented bounding box (OBB), and mask generation tasks, using a text-mask technique for pixel-level output. GeoPix [24] extends image-level tasks, such as captioning and visual question answering, to pixel-level segmentation, facilitating multireferring segmentation tasks. However, it requires users to provide target descriptions, increasing the interaction burden. GeoPixel [27] is the first end-to-end high-resolution RS LLM supporting pixel-level grounding, generating interleaved masks for fine-grained visual perception. Despite these advances, they largely rely on generic prompt templates, which are insufficient for capturing user-specific requirements in complex, object-dense scenes. Moreover, most existing frameworks overlook fine-grained interobject relationships and struggle to disambiguate visually similar targets, which are particularly prevalent in RS imagery. Distinct from prior approaches, our framework incorporates compact visual cues (bounding boxes) to guide model attention toward user-selected regions. Furthermore, by explicitly modeling interobject relationships and introducing semantic–relational alignment, our method bridges the gap between user intent and multimodal outputs, enabling precise, controllable segmentation and captioning in complex RS scenarios.

III. Method

We first introduce an overview of the CLV-Net framework in Section III-A, outlining its components and their interactions. In Sections III-B and III-C, we detail the VPReasoner and CMDecoder modules, which are designed to generate region-specific captions and refine cross-modal correlations for precise mask generation. Finally, Section III-D focuses on the semantic and relationship alignment (SRAlign) module, discussing its role in enforcing fine-grained semantic contrastive learning and ensuring relationship consistency between visual and textual representations.

A. Overview

Given an image $\boldsymbol{I} \in \mathbb{R}^{H\times W\times 3}$ with height H and width W, along with a user-provided prompt P_o, such as a box bounding a region of interest containing a target set O, CLV-Net aims to generate a corresponding pair of caption S and segmentation masks M that precisely aligns with the user's intention. Here, S describes the semantic information of the targets in O, and M represents the corresponding masks for them. To achieve this goal, we design a novel CLV-Net, which comprises three main components: a VPReasoner, a CMDecoder, and a SRAlign module. As illustrated in Fig. 2, our approach first employs the VPReasoner to generate a caption on the condition of the user-specified region. Next, the CMDecoder deeply explores cross-modal correlations between objects, yielding precise mask results. Finally, the SRAlign module enhances fine-grained discrimination between visually similar targets through the cross-modal semantic consistency loss, while maintaining visual–textual relationship consistency via the relationship consistency loss.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

B. Visual-Prompt Scene Reasoner

Given a global image $\boldsymbol{I}$ and a bounding box B specifying the region of interest I_r by a user, the VPReasoner generates a textual description S that faithfully meets the user's intention. Specifically, we employ ViT-H/14 CLIP [55] as the visual encoder $\mathcal{E}_v$ to extract global visual features f_x from $\boldsymbol{I}$. Furthermore, we utilize a pretrained object detector $\mathcal{E}_d$ [56], [57] to extract N_r object features $f_r \in \mathbb{R}^{N_r \times D} = \{O_i\}_{i=1}^{N_r}$ from I_r, along with their corresponding spatial coordinates $\{b_i\}_{i=1}^{N_r}$. These visual features are then mapped into the language space through a two-layer MLP, serving as a vision-to-language projection module ϕ_l, which shares its weights across both global and object features to ensure a consistent semantic mapping

$$f_x = \phi_l\left(\mathcal{E}_v\left(I\right)\right),\ f_r = \phi_l\left(\mathcal{E}_d\left(I_r\right)\right). \tag{1}$$

Meanwhile, we formulate a text query x_t as "Could you provide a global description of the image and a detailed description centered around the user-focused region <BOX>, and indicate the corresponding object phrases that appear in the generated description?" This query is then passed to a text encoder $\mathcal{E}_t$ to obtain a discrete embedding $f_t \in \mathbb{R}^{L \times D}$, where L represents the length of the text tokens. The formula is as follows:

$$f_t = \mathcal{E}_t\left(x_t\right). \tag{2}$$

Here, f_t contains natural language tokens and special placeholders <BOX>. The function Embed(f_t, f_r) directly substitutes the <BOX> placeholders at their corresponding positions with the visual object features f_r. This straightforward replacement and concatenation yields the fused prompt embedding $f_p \in \mathbb{R}^{N \times D}$, seamlessly integrating localized visual information into the textual sequence

$$f_p = \text{Embed}\left(f_t, f_r\right). \tag{3}$$

Subsequently, the global visual features f_x and the fused prompt embeddings f_p are concatenated along the sequence dimension to form a unified multimodal input [25], [27]. This sequence is then fed into the LLM $\mathcal{L}$ to generate a caption S, simultaneously yielding the object representation $f_h \in \mathbb{R}^{N \times D}$ containing N objects in S represented as a D-dimensional vector for each object

$$S, f_h = \mathcal{L}\left(\left[f_x; f_p\right]\right) \tag{4}$$

where $[\cdot;\cdot]$ denotes the concatenation operation along the sequence dimension. This concatenation-based fusion strategy enables the LLM's multihead self-attention (MHSA) mechanism to effectively integrate global and local information. The model utilizes the local feature f_r as a visual anchor to identify the target while attending to global features f_x for contextual cues. In the implementation, we extract f_h from the last transformer layer in the LLM.

Finally, we train the VPReasoner by using a captioning loss $\mathcal{L}_{\text{caption}}$, which is formulated as

$$\mathcal{L}_{\text{caption}} = \text{CE}\left(\hat{S}, S\right) \tag{5}$$

where CE represents the cross-entropy operation and $\hat{S}$ denotes the ground-truth textual labels. To efficiently adapt the LLM to our task, we employ low-rank adaptation (LoRA) [58] for parameter-efficient fine-tuning.

Instead of directly adopting LLMs to generate captions on global images, our method enables users to point out a region of interest to generate precise and context-relevant captions. Technically, we leverage object features as prompts to help the LLM in generating precise, intent-aligned captions, while simultaneously producing object-level textual features to support the following instance mask generation and fine-grained cross-modal alignment.

Notably, we employ HBBs rather than OBBs as visual prompts. This design serves a dual purpose: it minimizes user interaction costs by eliminating the need for orientation annotation, and it treats the background—often considered noise in detection tasks—as vital semantic context essential for scene reasoning. This visual-prompt-guided interaction reduces user effort and effectively mitigates the difficulties posed by the wide spatial coverage and high resolution of RS imagery, which otherwise make it hard to specify targets or regions using brief textual instructions. In contrast to region-based methods [41], [42] that restrict outputs to the interior of the user box and overlook salient cues from surrounding context, our approach integrates both in-box evidence and global scene information to yield more accurate and context-aware results.

C. Context-Aware Mask Decoder

Objects in RS images often exhibit visual indistinguishability, leading to misalignments between textual referring expressions and visual instances during mask generation. For example, in Fig. 1(a), two cars located on the upper left shore are mistakenly identified as boats, likely due to visual interference from the nearby boat on the right. To mitigate such errors, we propose a CMDecoder designed to disentangle visually similar objects and enhance cross-modal alignment. The CMDecoder consists of a context-aware graph former (CGFormer) and a mask decoder. By explicitly exploring the correlation among objects and integrating these correlations into the feature representation, CMDecoder effectively distinguishes visually similar objects and enhances the correspondence accuracy between textual phrases and visual objects.

Concretely, CGFormer first deeply explores cross-modal associations between the text-level object features f_h and the visual features f_r, yielding an enhanced text-level object feature $f_o \in \mathbb{R}^{N \times D}$ that accurately captures the cross-modal properties of the referenced targets from both modalities

$$f_o = \phi_o\left(f_h + \text{MHCA}\left(f_h, f_r, f_r\right)\right) \tag{6}$$

where MHCA represents the multihead cross-attention layers, and ϕ_o denotes a linear transformation. Next, our approach constructs a relationship matrix $\mathcal{R}_e \in \mathbb{R}^{N \times N}$ to explicitly model the interobject dependencies, which is formulated as follows:

$$\mathcal{R}_e = \text{Norm}\left(\phi_h\left(f_o\right) \cdot \phi_v\left(f_o^{\top}\right)\right) \tag{7}$$

where ϕ_h and ϕ_v represent two linear transformations, and *Norm* is a normalization operation to calculate normalized

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

relation weights on each object. The high value r_{ij} in $\mathcal{R}_e$ indicates the strong correlation between the objects i and j. We then fuse this object correlation information into f_o to generate a relationship-augmented object feature as

$$f_e = \phi_e\left((\mathcal{R}_e + \mathrm{I})\, f_o\right) \tag{8}$$

where I is the identity matrix for skip connections. Here, $(\mathcal{R}_e + I)f_o$ represents the feature aggregation based on the relation matrix, and $\phi_e(\cdot)$ denotes a linear transformation that projects the aggregated features into the target embedding space. We observe that the relation matrix $\mathcal{R}_e$ contains both meaningful connections and spurious, low-magnitude associations. To suppress weak or noisy relations while adapting to varying scene densities, we derive a binary object adjacency matrix A_o via a thresholding strategy. Specifically, for each object i, we rank its relation scores with all other objects $\{e_{i,j} \mid 1 \le j \le N\}$ in descending order. We then define the dynamic threshold as $\tau = e^{(k)}$, where $e^{(k)}$ corresponds to the kth largest relation score for object i. Subsequently, we set $A_o(i, j) = 1$ if the relation score $e_{i,j} \ge \tau$, and 0 otherwise. This procedure effectively constructs a K-nearest neighbor (KNN) graph, ensuring that each object retains its most salient interobject dependencies while filtering out weak or noisy associations, regardless of the global scene density. Subsequently, we leverage the refined adjacency matrix A_o to integrate the relation-enhanced features f_e into the object node representations f_o. This yields the final context-aware object representations $f_g \in \mathbb{R}^{N\times D}$, which encode both visual semantics and robust relational cues. This process is expressed as follows:

$$f_g = \psi\left(f_o, \phi_g\left(f_e \cdot A_o\right)\right) \tag{9}$$

where $\psi(,)$ indicates the element-wise addition of two components followed by an MLP transformation, and ϕ_g is an MLP operation layer.

For mask generation, we use the SAM2 decoder [59] to output masks. The decoder $\mathcal{D}$ takes as input the global visual feature f_x, and the context-aware object features f_g to produce binary region masks $M \in \mathbb{R}^{N\times H\times W}$. The mask loss $\mathcal{L}_{\text{Mask}}$ combines Dice loss and binary cross-entropy (BCE) loss to optimize the mask prediction.

Unlike previous methods that directly decode masks using text features, our approach leverages the CMDecoder to deeply mine cross-modal associations between textual and visual cues, while simultaneously capturing semantic relationships among different targets. This enables the generation of more discriminative referring features, effectively addressing the challenges commonly encountered in RS scenes. By capturing these nuanced associations, our approach reduces mismatches between textual descriptions and visual targets, leading to more accurate alignment and segmentation.

D. Semantic and Relationship Consistency Learning

To improve both textual and visual decoding accuracy and consistency, we introduce the SRAlign, which facilitates cross-modal alignment at the level of object semantics and inter-object relationships, ensuring effective integration of valuable textual and visual information. As illustrated in Fig. 3, noun

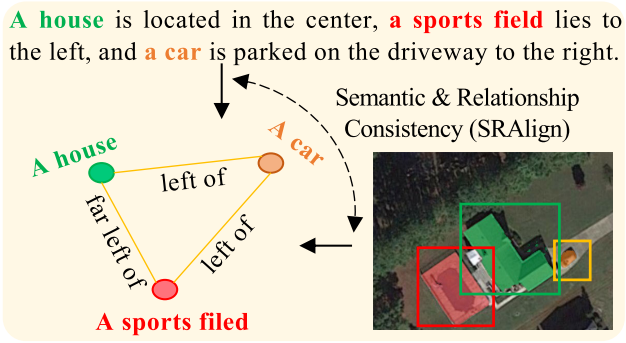


Fig. 3. Semantic and relationship alignment (SRAlign). SRAlign optimizes cross-modal alignment by encouraging entity-level semantic consistency and relationship-level consistency, yielding captions and masks that are mutually faithful and coherent across modalities.

phrases in the generated caption should correspond to their segmentation masks in the image (e.g., the phrase "a house" aligns with the green mask covering the house), and the relationships described in text should mirror the spatial or semantic relationships among the segmented objects (e.g., "a house is located between the sports field on the left and the car on the right" should be reflected by the relative positions of the corresponding masks). Motivated by these observations, SRAlign comprises two complementary objectives: a semantic consistency loss $\mathcal{L}_{\text{csc}}$, which promotes fine-grained semantic discrimination across modalities, and a relationship consistency loss $\mathcal{L}_{\text{rec}}$, which aligns meaningful commonalities between textual relationships and visual interactions. Together, these losses encourage precise, interpretable correspondences between caption tokens and segmentation masks.

1) Cross-Modal Semantic Consistency Loss: Given a predicted mask-caption pair (M, S), it is important to note that the relationship between masks and words is not strictly one-to-one. Therefore, the ith mask embedding $\mathbf{m}_i$ is expected to be close to the matched word embedding sets $\mathbf{s}_i{}^+$, while maintaining a significant distance from other word embeddings $\mathbf{s}_j$. In other words, the mutual information between mask entities and their corresponding words should be maximized. However, directly optimizing mutual information is computationally infeasible due to its high cost and complexity. To address this, we introduce a cross-modal semantic consistency loss $\mathcal{L}_{\text{csc}}$ inspired by InfoNCE [60], which provides a variational lower bound for mutual information optimization. Unlike InfoNCE, typically used for unimodal instance discrimination, we adapt it to our cross-modal scenario by explicitly aligning the visual and textual domains in a bidirectional manner. We extend the original one-to-one anchor-positive mapping to accommodate one-to-many or many-to-many relationships, aggregating the contrastive penalty over valid positive matches for our dense reasoning task. This loss is formulated as follows:

$$\begin{aligned}\mathcal{L}_{\text{csc}} = &-\sum_{i=1}^{N}\frac{1}{\left|\{\mathbf{s}_i\}^+\right|}\sum_{\mathbf{s}_i\in\{\mathbf{s}_i\}^+}\log\frac{\exp\left(\mathbf{m}_i^\top\mathbf{s}_i/\tau\right)}{\sum_{j\neq i}\exp\left(\mathbf{m}_i^\top\mathbf{s}_j/\tau\right)}\\ &-\sum_{i=1}^{N}\frac{1}{\left|\{\mathbf{m}_i\}^+\right|}\sum_{\mathbf{m}_i\in\{\mathbf{m}_i\}^+}\log\frac{\exp\left(\mathbf{s}_i^\top\mathbf{m}_i/\tau\right)}{\sum_{j\neq i}\exp\left(\mathbf{s}_i^\top\mathbf{m}_j/\tau\right)}\end{aligned} \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

where τ is a learnable parameter. Likewise, we impose that the ith word embedding $\mathbf{s}_i$ is similar to the matched mask embedding sets $\{\mathbf{m}_i\}^+$ and dissimilar to the irrelevant $\mathbf{m}_j$.

2) Relationship Consistency Loss: The relationship consistency loss is designed to align relational structures across modalities by identifying meaningful commonalities, ensuring consistency between object relationships in the visual domain and their corresponding relationships in the textual embedding space. Rather than enforcing a strict one-to-one mapping, it aligns shared relational distributions, filtering noise while preserving valid cross-modal correlations. Given a predicted mask–caption pair (M, S), we first extract the textual object representation f_h^t from the last transformer layer of the LLM and compute the textual relationship matrix $\mathcal{R}_e^t$ for the objects referenced in (S) following (6) and (7). Similarly, we obtain the visual object representation f_h^v from a hidden layer of the SAM2 decoder and construct the corresponding visual relationship matrix $\mathcal{R}_e^v$ for the objects segmented in M using the same procedure. To encourage the model to learn consistent object relationships between the visual space and their counterparts in the textual embedding space, we minimize the Kullback–Leibler divergence between the two relationship distributions

$$\begin{aligned}\mathcal{L}_{\text{rec}} &= \tfrac{1}{2}\left[\text{KL}\left(\mathcal{R}_e^t \| \mathcal{R}_e^v\right) + \text{KL}\left(\mathcal{R}_e^v \| \mathcal{R}_e^t\right)\right] \\ &= \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\left[\mathcal{R}_{\text{e}}^t(i,j)\ \log\frac{\mathcal{R}_{\text{e}}^t(i,j)}{\mathcal{R}_{\text{e}}^v(i,j)}\right. \\ &\qquad \left. + \mathcal{R}_{\text{e}}^v(i,j)\ \log\frac{\mathcal{R}_{\text{e}}^v(i,j)}{\mathcal{R}_{\text{e}}^t(i,j)}\right]. \end{aligned} \tag{11}$$

The SRAlign loss $\mathcal{L}_{\text{SRAlign}}$ is formulated as

$$\mathcal{L}_{\text{SRAlign}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{csc}}. \tag{12}$$

The minimization of $\mathcal{L}_{\text{SRAlign}}$ optimizes the alignment between corresponding mask-caption pairs by ensuring consistency in both semantic and relational information across modalities. This joint optimization enhances the model's ability to generate captions and masks that are not only accurate but also semantically and relationally aligned, thereby producing more coherent and consistent cross-modal outputs.

Finally, we train our model using either $\mathcal{L}_{\text{Caption}}$ or $\mathcal{L}_{\text{Mask}}$, combined with the proposed $\mathcal{L}_{\text{SRAlign}}$. The total loss is calculated as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Caption}} + \mathcal{L}_{\text{Mask}} + \lambda\mathcal{L}_{\text{SRAlign}} \tag{13}$$

where λ is a balancing weight.

IV. Experiments

A. Datasets and Metrics

We conduct comprehensive experiments on both the RS dataset GeoPixelD and the natural-image dataset GranD. Specifically, GeoPixelD is employed to evaluate the controllable and relation-aware consistency of our method on high-resolution and object-dense RS imagery, while GranD further demonstrates the generalizability of our approach to diverse natural-image domains.

GeoPixelD is a multimodal grounded conversation generation (GCG) dataset comprising 53 816 grounded phrases aligned with 600 817 object masks, specifically designed for RS image understanding. It provides hierarchically structured annotations that deliver rich semantic descriptions, seamlessly integrating comprehensive scene-level context with fine-grained, localized object-level details.

GranD is explicitly designed for the GCG task in natural-image domains and comprises approximately 214 000 image–grounded text pairs. Of these samples, 2600 are reserved for validation and 5000 for testing, with the remaining examples serving as training data to support large-scale model development and evaluation.

We evaluate our model using METEOR and CIDEr to assess caption quality, class-agnostic mask average precision (AP) to measure mask-to-phrase alignment, mask IoU to quantify mask quality, and mask recall to evaluate region-specific grounding performance.

B. Implementation Details

1) Experimental Settings: We adopt InternLM2.5-7B [61] as the LLM for caption generation and employ LoRA [58] for efficient fine-tuning. A fixed CLIP-ViT-L/14-336 [55] serves as the visual encoder, complemented by a grounded vision encoder initialized from SAM2 [59] weights, while mask generation is performed by the SAM2 decoder. The trainable components of our architecture include the mask decoder ($\mathcal{D}$), LoRA parameters ($\alpha = 8$), and CGFormer, where the MHCA mechanism uses eight attention heads with a 0.1 dropout rate. The CMDecoder uses a dynamic thresholding strategy $\tau = e^{(k)}$ with $k = 5$. In (13), we set $\lambda = 1$ to balance the optimization objectives. The learning rate is linearly warmed up to a maximum of 3×10^{-4} over the first 150 training steps and then gradually decreased following a cosine decay schedule. The model is trained for 10 epochs with a batch size of 8 using 4 NVIDIA A6000 GPUs.

2) Efficient Pair Construction for Semantic Consistency: To ensure high training efficiency, even in dense scenes with a large number of objects N, we employ a linear complexity sampling strategy for the semantic consistency loss. Specifically, positive pairs are directly retrieved from ground-truth visual-text alignments, while negative pairs are randomly selected. To maintain a balanced ratio between positive and negative pairs, we select an equal number of negative pairs. This strategy ensures that both computational cost and memory usage scale linearly ($O(N)$) with the number of objects, minimizing overhead while preserving the ability to learn discriminative features. During testing, no pair construction is required, further reducing computational demands.

C. Comparison With State-of-the-Art Methods

We conduct comprehensive evaluations of our approach against leading state-of-the-art methods, including LISA [20], PixelLM [38], GLaMM [22], and GeoPixel [27], on both the RS dataset GeoPixelD and the natural-image dataset GranD. As summarized in Table I, models are assessed using multiple metrics, including CIDEr, METEOR, AP50, mIoU, and recall.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

TABLE I

TO VALIDATE THE EFFECTIVENESS AND GENERALIZATION OF OUR METHOD, WE PROVIDE A PERFORMANCE COMPARISON WITH SEVERAL ADVANCED METHODS ON THE GCG TASK USING THE RS DATASET GEOPIXELD AND THE NATURAL-IMAGE DATASET GRAND. † DENOTES MODELS THAT WERE FINE-TUNED ON THE TRAINING DATA. CLV-NET OUTPERFORMS ALL OTHER MODELS ACROSS ALL METRICS

Model	GeoPixelD					GranD					Time		
	CIDEr	METEOR	AP50	MIoU	RECALL	CIDEr	METEOR	AP50	MIoU	RECALL	Captioning (s/img)	Segmentation (s/img)	Total (FPS)
LISA-7B† (CVPR'24)	14.6	22.3	8.5	42.7	29.0	33.9	13.0	25.2	62.0	36.3	1.52	0.13	0.61
PixelLM-7B† (CVPR'24)	18.3	22.5	10.5	42.4	29.6	36.5	15.4	28.1	65.6	37.2	29.25	0.11	0.03
GLaMM-7B† (CVPR'24)	15.7	23.0	12.5	46.4	32.8	**47.2**	16.2	30.8	66.3	41.8	11.32	0.10	0.09
GeoPixel-7B (ICML'25)	21.6	24.0	19.0	52.3	38.8	18.7	10.5	21.7	49.8	32.4	21.45	0.16	0.05
CLV-Net-7B (ours)	**24.5**	**26.2**	**21.8**	**55.3**	**41.9**	47.0	**16.8**	**33.4**	**69.2**	**43.6**	11.86	0.09	0.08

Our method consistently outperforms existing approaches, demonstrating robust performance and superior generalization across diverse domains.

As shown in Table I, methods designed specifically for natural-image understanding exhibit clear limitations when applied to RS data. LISA† performs worst on the RS benchmarks due to its architectural limitations and its reliance on simple scene descriptions and coarse instance-level segmentation, which are poorly suited to RS imagery. PixelLM† achieves moderate improvements over LISA† by incorporating multiscale feature scaling and semantic token partitioning. GLaMM† benefits from decoupled regional and global visual encodings and a unified multitask pretraining strategy, which yields improvements over earlier baselines, but it does not explicitly address the distinct challenges posed by RS data and therefore remains outperformed by GeoPixel and our method. GeoPixel, which is specifically tailored for RS imagery, achieves substantial improvements compared with previous general-purpose models. Nevertheless, it still lags behind our approach. The performance gap is even larger on the natural-image dataset, further highlighting the strong cross-domain generalization of our method. Notably, we observe that on RS datasets, natural-image-oriented models, even after fine-tuning, exhibit a significant performance gap compared with specialized RS models. Conversely, on natural-image datasets, GeoPixel underperforms compared with methods originally designed for natural scene understanding. In contrast, our approach not only achieves the best performance on the RS benchmark but also reaches comparable or even superior results on natural-image datasets. This advantage stems primarily from the ability of our method to capture fine-grained object semantics and to model relationships across modalities. Moreover, the model produces both globally coherent representations and finely detailed local outputs that align with user intent, thereby enhancing interpretable multimodal reasoning across domains.

D. Qualitative Results

Fig. 4 presents the visualization results of our method compared to GeoPixel [27]. The yellow underline in the text corresponds to the yellow dashed regions in the figure, indicating areas where the segmentation results fail to capture the targets described in the text. Cross marks in the image indicate incorrect matches between the segmented targets and the textual descriptions. In Fig. 4(a), GeoPixel generates a coarse, holistic description along with corresponding target masks, but it fails to capture most of the regional details and lacks a deep understanding of the scene. Within the yellow dashed regions, although the text specifies certain targets, the visual localization fails to identify them. In the red dashed regions, the text-related targets are located, but the segmentation quality is poor. In particular, the regions marked with crosses show incorrect matches between the target masks and the textual descriptions. These shortcomings significantly hinder the user's ability to accurately interpret the content of the image. In contrast, as shown in Fig. 4(b), our method generates a global description, detailed local descriptions corresponding to regions of user interest, and high-quality target masks. The segmented masks exhibit precise boundaries, and the noun entities in the captions are accurately aligned with the corresponding objects in the image. In addition, we evaluate the models' ability to handle visually similar targets. For example, in the first column of Fig. 4(a), GeoPixel misclassifies a truck as a "boat" due to visual ambiguity, whereas CLV-Net correctly identifies the truck by considering the global spatial context (e.g., the surrounding road). In scenes with densely packed, visually similar objects, such as baseball fields (Col. 3), aircraft (Col. 4), and docked boats (Col. 5), GeoPixel struggles with separating instances, leading to missed or incorrect detections. CLV-Net, however, effectively disentangles these targets by integrating both visual and textual semantic information, successfully disambiguating visually similar objects based on their unique spatial and semantic contexts.

Overall, our approach leverages visual-prompt-guided interaction to reduce user effort and effectively mitigate the challenges posed by the wide spatial coverage and high resolution of RS imagery. Unlike region-based methods that constrain outputs to the interior of the user-defined box and ignore contextual cues from the surrounding area, our method integrates both in-box evidence and global scene information to produce more accurate and contextually aware results.

E. Analysis of VPReasoner

Our VPReasoner integrates local visual information with global features via a visual-prompt box, guiding the LLM to generate both a global description and fine-grained local details of interest to the user. To quantitatively evaluate the effectiveness of visual-prompt guidance, we conduct a comprehensive analysis examining the impact of visual-prompt guidance on both our method and baseline models (see Table II), alongside an assessment of model scaling effects (see Table III).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

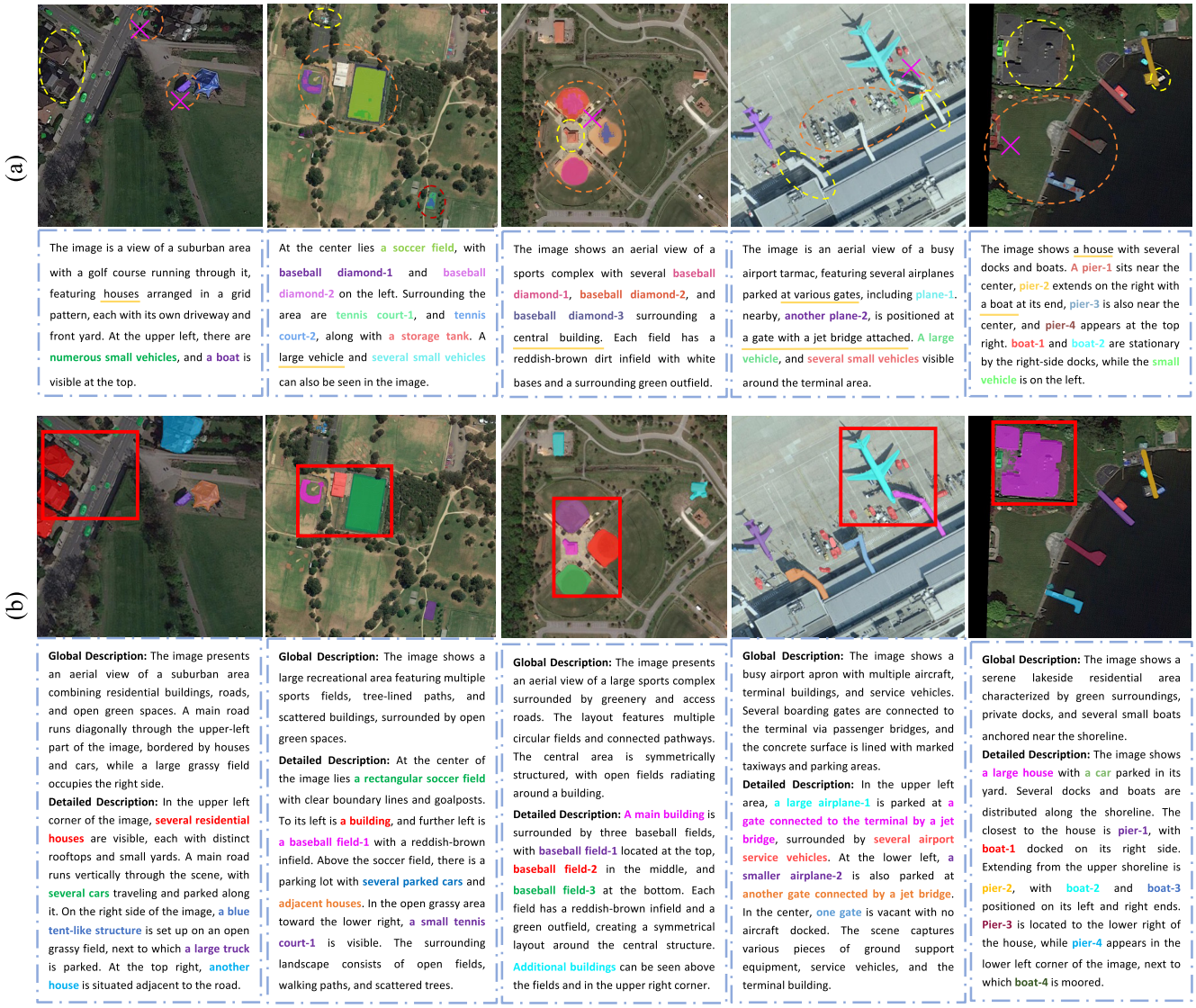


Fig. 4. Quantitative results of our(a) GeoPixel and (b) CLV-Net on the GeoPixelD dataset, where the colors indicate the matching relationship between masks and words. Underlined text denotes predicted words with missing masks, while crosses indicate mismatches between the generated text and the target.

TABLE II

EVALUATION OF THE IMPACT OF ADAPTED BASELINES AND THE REMOVAL OF THE VISUAL-PROMPT BOX ON THE GEOPIXELD DATASET

Method	CIDEr	METEOR	AP50	MIoU	RECALL
Without Prompt Box					
PixelLM	18.3	22.5	10.5	42.4	29.6
GeoPixel	15.7	23.0	12.5	46.4	32.8
CLV-Net	**22.7**	**25.0**	**19.2**	**53.5**	**39.6**
With Prompt Box					
PixelLM	19.1	23.2	13.6	45.5	32.7
GeoPixel	17.3	23.8	16.9	48.0	34.6
CLV-Net	**24.5**	**26.2**	**21.8**	**55.3**	**41.9**

TABLE III

EVALUATION OF THE IMPACT OF LLM SCALE

Method	CIDEr	METEOR	AP50	MIoU	RECALL
CLV-Net (7B)	**24.5**	**26.2**	**21.8**	**55.3**	**41.9**
Using Different LLMs Size					
CLV-Net (1.8B)	23.8	25.6	21.4	55.1	41.3

1) Impact of Visual-Prompt Guidance: Because our task emphasizes controllable outputs, the system must accept a user-provided visual prompt and then produce the user-desired results, thereby ensuring alignment with user intent. As shown in Table II (comparing "with prompt box" versus "without prompt box"), removing the prompt box results in a marked decline in both captioning and segmentation metrics. Specifically, in the "without prompt box" setting, the regional object feature f_r is omitted and replaced by the global visual feature f_x, causing the model to default to processing the entire image. For CLV-Net, the absence of explicit spatial guidance causes the model to revert to coarse-grained scene modeling, often leading to the hallucination of irrelevant background objects or the omission of specific targets. To isolate the contribution of our architectural design from the benefits of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

PERFORMANCE ANALYSIS COMPARING THE IMPACT OF DIFFERENT CMDECODER COMPONENTS ON CLV-NET PERFORMANCE ON THE GEOPIXELD DATASET, INCLUDING THE EFFECT OF REMOVING CGFORMER (W/O CGFORMER) AND USING DIFFERENT MASK DECODERS

Method	CIDEr	METEOR	AP50	MIoU	RECALL
CLV-Net (SAM2)	**24.5**	**26.2**	**21.8**	**55.3**	**41.9**
Without CGFormer					
CLV-Net (w/o CGFormer)	23.2	25.6	20.5	53.0	39.4
Using Different Mask Decoders					
CLV-Net (SAM)	24.4	26.0	21.2	54.8	41.5

the spatial prior, we adapted the baseline models, PixelLM and GeoPixel, to accept the same bounding box inputs ("with prompt box" in Table II). Even with identical visual prompts, CLV-Net significantly outperforms both adapted baselines. Specifically, CLV-Net surpasses the prompt-adapted GeoPixel, demonstrating that the performance improvements stem from the effective global–local feature fusion and context-aware alignment within the VPReasoner and CMDecoder, rather than solely the introduction of the visual prompt.

2) Impact of LLM Scale: We further evaluated the impact of replacing the InternLM2.5-7B backbone with InternLM2.5-1.8B, and report the results in Table III ("Using Different LLMs"). Replacing the 7B model with the 1.8B variant yields a modest drop in captioning metrics while segmentation performance remains comparable. The performance change associated with model scale is substantially smaller than the degradation observed when the visual prompt is removed. These results indicate that the primary source of our performance gains arises from the newly proposed modules rather than from backbone model size.

F. Analysis of CMDecoder

Our CMDecoder employs a CGFormer to model interobject dependencies and to bridge the visual and textual semantic gap, thereby producing enriched object representations. These enhanced representations are then passed to a mask decoder to generate the corresponding segmentation outputs. To quantitatively evaluate the effectiveness of the CMDecoder, we perform a comprehensive ablation study that considers both the removal of CGFormer (w/o CGFormer) and the use of different mask decoders, as reported in Table IV.

Effect of removing CGFormer. The CGFormer in the CMDecoder explicitly captures relationships among objects and aligns visual features with textual semantics to improve segmentation fidelity and cross-modal consistency. As shown in Table IV (w/o CGFormer), removing the CGFormer causes declines in both captioning and segmentation metrics, with a larger drop observed in segmentation. This result indicates that richer, context-aware object representations substantially benefit visual decoding, and that optimizing these representations jointly also positively influences caption generation.

Using different mask decoders. To assess the impact of the pretrained SAM2 model within our framework, we replace SAM2 with the SAM. As reported in Tables I and IV, the model equipped with SAM still outperforms the baselines. Although performance shows a modest decrease relative to the SAM2-based configuration, these results further confirm the effectiveness and generalizability of our framework.

TABLE V

ABLATION STUDY ON THE HYPERPARAMETER k. EFFECT OF FILTERING THE ADJACENCY MATRIX TO RETAIN VARYING NUMBERS OF THE MOST SALIENT DEPENDENCIES

k	AP50	MIoU	RECALL
0	19.5	54.2	40.3
3	20.9	54.6	41.2
5	**21.8**	**55.3**	**41.9**
8	21.6	55.0	41.5
10	21.1	54.7	41.3

TABLE VI

PERFORMANCE COMPARISON OF THE IMPACT OF REMOVING BOTH OR EITHER OF THE LOSSES, $\mathcal{L}_{\text{REC}}$ AND $\mathcal{L}_{\text{csc}}$, IN $\mathcal{L}_{\text{SRALIGN}}$ ON CLV-NET PERFORMANCE ON THE GEOPIXELD DATASET

L_{csc}	L_{rec}	CIDEr	METEOR	AP50	MIoU	RECALL
-	-	23.7	25.3	19.6	53.8	40.2
✓	-	24.2	25.6	21.5	55.1	41.3
-	✓	24.0	25.8	21.3	54.9	41.0
✓	✓	**24.5**	**26.2**	**21.8**	**55.3**	**41.9**

Sensitivity analysis of relation sparsity (k). We explored the impact of the neighbor count k on model performance. As shown in Table V, the model achieves peak performance at $k = 5$. This setting effectively balances the preservation of semantic context with the suppression of noise. Lower values ($k < 5$) fail to capture sufficient dependencies, while higher values ($k > 5$) introduce spurious correlations, degrading the AP50 and MIoU scores.

G. Analysis of SRAlign

We conducted an ablation study to validate the effectiveness of the SRAlign loss $\mathcal{L}_{\text{SRAlign}}$, which comprises two components, $\mathcal{L}_{\text{csc}}$ and $\mathcal{L}_{\text{rec}}$ [refer to (10)–(12)]. The results are summarized in Table VI. Both $\mathcal{L}_{\text{csc}}$ and $\mathcal{L}_{\text{rec}}$ yield substantial improvements across captioning and segmentation metrics. Specifically, $\mathcal{L}_{\text{csc}}$ enforces object-level semantic alignment between predicted masks and their corresponding caption words, thereby establishing a robust cross-modal correspondence that enriches the semantic representation of objects and sharpens distinctions among different targets. Meanwhile, $\mathcal{L}_{\text{rec}}$ models interobject relationships and captures valuable dependency cues among targets. By learning consistency of relationships across modalities, $\mathcal{L}_{\text{rec}}$ promotes mutual reinforcement between textual and visual decoding. Joint optimization of these two components further enhances model performance, demonstrating the efficacy of the proposed $\mathcal{L}_{\text{SRAlign}}$.

Additional evidence is provided in Table VII, which examines the overall effectiveness of $\mathcal{L}_{\text{SRAlign}}$ under varying values of λ in (13). Optimal performance is achieved when $\lambda = 1$. We observe a differential impact of λ on the two tasks. Specifically, when $\lambda = 0$ (i.e., abating the alignment loss), the degradation in segmentation metrics (e.g., AP50 drops by 2.2 points) is more pronounced than in captioning metrics (e.g., CIDEr drops by 0.8 points). We attribute this phenomenon to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

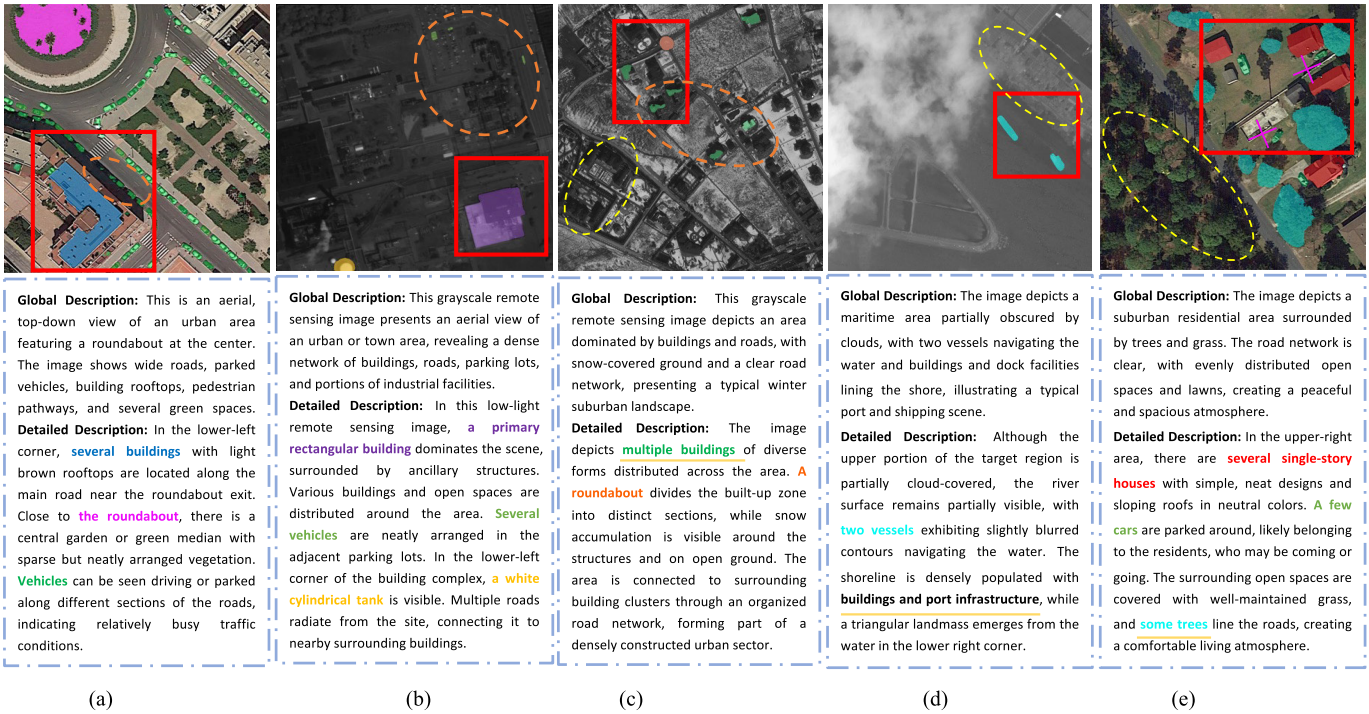


Fig. 5. Visualization of several failure cases, where the colors indicate the matching relationship between masks and words. Red rectangles denote user-provided visual prompts; underlined text and yellow dashed circles denote predicted text with missing masks, while orange dashed circles indicate low-quality mask segmentation. (a) Lighting variations. (b) Low light. (c) Background interference. (d) Cloud occlusion. (e) MLLM hallucinations.

TABLE VII
ABLATION STUDY ON THE IMPACT OF DIFFERENT WEIGHTINGS OF THE $\mathcal{L}_{\text{SRALIGN}}$ ON THE PERFORMANCE OF CLV-NET

λ	CIDEr	METEOR	AP50	MIoU	RECALL
0	23.7	25.3	19.6	53.8	40.2
0.5	24.3	25.9	21.5	55.0	41.4
1	**24.5**	**26.2**	**21.8**	**55.3**	**41.9**
5	23.9	25.4	21.2	54.6	41.1

inherent differences in the model components. The caption generation is supported by the robust, pretrained knowledge of the MLLM, making it relatively resilient to weight fluctuations. Conversely, the CMDecoder is a lighter, task-specific module that relies heavily on the explicit supervision provided by $\mathcal{L}_{\text{SRAlign}}$ to learn precise cross-modal associations and boundary delineations.

H. Failure Analysis

To provide a comprehensive assessment of model limitations, we visualize representative failure cases in Fig. 5, categorizing them into perception-level constraints and generative hallucinations.

1) Perception-Level Failures: Our method produces imprecise segmentation contours in scenes where targets are obscured by shadows, lighting variations, or blur, as shown in Fig. 5(a). In low-illumination scenes or those with strong background clutter, targets exhibit low contrast relative to their surroundings. This reduces feature discriminability, leading to incomplete or missed segmentations [Fig. 5(b)]. Furthermore, extreme weather conditions, such as snow or heavy fog, can obscure salient features, causing the model to fail in segmenting the targets described in the text [Fig. 5(c) and (d)].

2) Generative Hallucinations: Beyond perceptual limitations, we observe hallucination phenomena characteristic of MLLMs. When visual evidence is weak or the target is difficult to recognize, the model tends to over-rely on strong parametric language priors rather than strictly adhering to visual cues. For instance, as shown in Fig. 5(e), the model encounters a boat parked on land near buildings. Due to the visual ambiguity of the target and the strong semantic association between "buildings" and "cars" in the language prior, the model hallucinates the object as a "car," favoring semantic probability over visual reality. This highlights the ongoing challenge of balancing linguistic knowledge with strict visual fidelity in MLLMs.

I. Inference Speed and Model Parameters

In our CLV-Net framework, only the LoRA parameters and the CMDecoder are trainable, together accounting for merely 7.28% of the total model parameters. This lightweight design substantially reduces both training and storage overhead while preserving strong representational capacity across modalities. To further assess computational efficiency, we compare the inference speed of CLV-Net with several representative models, including LISA, PixelLM, GLaMM, and GeoPixel. As shown in Table I, CLV-Net achieves 0.08 FPS on an NVIDIA A6000 GPU, while LISA, PixelLM, GLaMM, and GeoPixel reach 0.61, 0.03, 0.09, and 0.05 FPS, respectively. While CLV-Net is slower than LISA, this is due to the inherent complexity

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

of multitarget reasoning. LISA, optimized for simpler single-target tasks, generates shorter descriptions, resulting in faster captioning latency (1.52 s). In contrast, CLV-Net produces longer, more detailed scene descriptions, adding computational load. Despite this, CLV-Net demonstrates superior efficiency among multitarget models. our captioning latency (11.86 s) is significantly lower than that of GeoPixel (21.45 s) or PixelLM (29.25 s). This confirms that the VPReasoner effectively streamlines the generation process by guiding the LLM toward targeted, efficient descriptions. We further observe that the autoregressive caption generation of the LLM remains the primary bottleneck across all models—a challenge that warrants future investigation. Overall, our approach requires relatively few learnable parameters and supports efficient inference, achieving a favorable balance between model complexity, computational cost, and multimodal reasoning performance.

J. Limitations and Future Perspectives

Despite the promising performance of CLV-Net, several limitations remain that warrant further exploration. When the input data exhibit sensor deficiencies or environmental disturbances such as fog, snow, or low-light noise, the model often produces incomplete or imprecise segmentations. Its performance also degrades in complex scenarios characterized by heavy occlusion, illumination variation, or densely distributed targets, revealing limited robustness under diverse sensing and environmental conditions. Future research will focus on enhancing the model's adaptability through multimodal RS data fusion, which can better capture complementary spatial and spectral information. In addition, we acknowledge that the current inference speed poses a challenge for real-time RS applications. The primary computational burden stems from the autoregressive generation of long captions, particularly in multitarget reasoning tasks. In future work, we aim to address this bottleneck by exploring lightweight network architectures and model quantization techniques to improve efficiency without compromising accuracy.

V. Conclusion

In this work, we propose CLV-Net, a framework designed to generate precise, user-tailored multimodal outputs by leveraging simple visual cues, such as bounding boxes, to guide attention toward regions of interest. CLV-Net integrates a CMDecoder to model and incorporate interobject semantic relationships, enhancing mask quality and target representations, and a SRAlign module that enforces cross-modal representation and relationship consistency while improving discrimination among visually similar targets through a combination of cross-modal semantic consistency loss and relationship consistency loss. Extensive experiments on both the RS dataset GeoPixelD and the natural-image dataset GranD demonstrate that CLV-Net consistently outperforms existing state-of-the-art methods. The framework effectively captures user intent, resolves fine-grained visual ambiguities, and produces high-quality, aligned caption and mask outputs across diverse visual domains.

References

[1] X. Dai, Y. Chen, J. Yang, P. Zhang, L. Yuan, and L. Zhang, "Dynamic DETR: End-to-end object detection with dynamic attention," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2968–2977.

[2] I. Gulrajani, T. Hashimoto, X. Li, P. Liang, and J. Thickstun, "Diffusion-LM improves controllable text generation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 4328–4343.

[3] J. Luo et al., "Semantic-conditional diffusion networks for image captioning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 23359–23368.

[4] J. Yu, J. Yao, J. Zhang, Z. Yu, and D. Tao, "SPRNet: Single-pixel reconstruction for one-stage instance segmentation," *IEEE Trans. Cybern.*, vol. 51, no. 4, pp. 1731–1742, Apr. 2021.

[5] J. Gao et al., "Segment everything everywhere all at once," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 19769–19782.

[6] H. Ren, M. Xia, L. Weng, H. Lin, J. Huang, and K. Hu, "Interactive and supervised dual-mode attention network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–18, 2025.

[7] X. Zhang, K. Yang, J. Lin, J. Yuan, Z. Li, and S. Li, "PVPUFormer: Probabilistic visual prompt unified transformer for interactive image segmentation," *IEEE Trans. Image Process.*, vol. 33, pp. 6455–6468, 2024.

[8] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[9] S. Li et al., "Bi-mapper: Holistic BEV semantic mapping for autonomous driving," *IEEE Robot. Autom. Lett.*, vol. 8, no. 11, pp. 7034–7041, Nov. 2023.

[10] J. Lin et al., "EchoTrack: Auditory referring multi-object tracking for autonomous driving," 2024, *arXiv:2402.18302*.

[11] Y. Wei et al., "Editable scene simulation for autonomous driving via collaborative LLM-agents," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 15077–15087.

[12] X. Liu et al., "A generalist medical language model for disease diagnosis assistance," *Nature Med.*, vol. 31, no. 3, pp. 932–942, Mar. 2025.

[13] H. Tang, J. Liu, S. Yan, R. Yan, Z. Li, and J. Tang, "M3Net: Multi-view encoding, matching, and fusion for few-shot fine-grained action recognition," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 1719–1728.

[14] H. Tang, C. Yuan, Z. Li, and J. Tang, "Learning attention-guided pyramidal features for few-shot fine-grained recognition," *Pattern Recognit.*, vol. 130, Oct. 2022, Art. no. 108792.

[15] H. Tang, S. He, and J. Qin, "Connecting giants: Synergistic knowledge transfer of large multimodal models for few-shot learning," in *Proc. 34rd Int. Joint Conf. Artif. Intell.*, Aug. 2025, pp. 6227–6235.

[16] H. Tang, S. He, and J. Qin, "Connecting giants: Synergistic knowledge transfer of large multimodal models for few-shot learning," in *Proc. 33rd Int. Joint Conf. Artif. Intell.*, Aug. 2024, pp. 6227–6235.

[17] H. Tang, Z. Li, Z. Peng, and J. Tang, "BlockMix: Meta regularization and self-calibrated inference for metric-based meta-learning," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 610–618.

[18] B. He et al., "MA-LMM: Memory-augmented large multimodal model for long-term video understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 13504–13514.

[19] Q. Wang, Z. Yang, W. Ni, J. Wu, and Q. Li, "Semantic-spatial collaborative perception network for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, 2024.

[20] X. Lai et al., "LISA: Reasoning segmentation via large language model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 9579–9589.

[21] Y. Zhang, Z. Ma, X. Gao, S. Shakiah, Q. Gao, and J. Chai, "Groundhog grounding large language models to holistic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 14227–14238.

[22] H. Rasheed et al., "GLaMM: Pixel grounding large multimodal model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 13009–13018.

[23] X. Zhang et al., "SGDiff: Scene graph guided diffusion model for image collaborative segcaptioning," in *Proc. AAAI*, 2025, pp. 10257–10265.

[24] R. Ou, Y. Hu, F. Zhang, J. Chen, and Y. Liu, "GeoPix: A multimodal large language model for pixel-level image understanding in remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 3, pp. 324–337, Sep. 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.

[25] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "GeoChat: Grounded large vision-language model for remote sensing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27831–27840.
[26] Y. Zhou et al., "GeoGround: A unified large vision-language model for remote sensing visual grounding," 2024, *arXiv:2411.11904*.
[27] A. Shabbir, M. Zumri, M. Bennamoun, F. S. Khan, and S. Khan, "GeoPixel: Pixel grounding large multimodal model in remote sensing," in *Proc. ICML*, 2025, pp. 54095–54111.
[28] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol.*, vol. 1. Minneapolis, MI, USA: Association for Computational Linguistics, Jun. 2019, pp. 4171–4186.
[29] W. Zhang, M. Cai, T. Zhang, Y. Zhuang, and X. Mao, "EarthGPT: A universal multimodal large language model for multisensor image comprehension in remote sensing domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, 2024.
[30] W.-L. Chiang et al., "Judging LLM-as-a-judge with MT-bench and chatbot arena," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 46595–46623.
[31] Y. J. Lee, C. Li, H. Liu, and Q. Wu, "Visual instruction tuning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 34892–34916.
[32] M. Zhang, F. Chen, and B. Li, "Multistep question-driven visual question answering for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023.
[33] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 19730–19742.
[34] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," 2023, *arXiv:2304.10592*.
[35] X. Chen et al., "VisionLLM: Large language model is also an open-ended decoder for vision-centric tasks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 61501–61513.
[36] Z. Xia, D. Han, Y. Han, X. Pan, S. Song, and G. Huang, "GSVA: Generalized segmentation via multimodal large language models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3858–3869.
[37] C. Ma, Y. Jiang, J. Wu, Z. Yuan, and X. Qi, "Groma: Localized visual tokenization for grounding multimodal large language models," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2024, pp. 417–435.
[38] Z. Ren et al., "PixelLM: Pixel reasoning with large multimodal model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26364–26373.
[39] R. Su, Q. Yu, and D. Xu, "STVGBert: A visual-linguistic transformer based framework for spatio-temporal video grounding," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 1513–1522.
[40] H. Fei et al., "OMG-LLaVA: Bridging image-level, object-level, pixel-level reasoning and understanding," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 71737–71767.
[41] T. Wang et al., "Caption anything: Interactive image description with diverse multimodal controls," 2023, *arXiv:2305.02677*.
[42] X. Huang et al., "Segment and caption anything," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 13405–13417.
[43] Z. Yang, Q. Li, Y. Yuan, and Q. Wang, "HCNet: Hierarchical feature aggregation and cross-modal feature alignment for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–11, 2024.
[44] Z. Yang, W. Zhang, Q. Li, W. Ni, J. Wu, and Q. Wang, "C^{2}Net: Road extraction via context perception and cross spatial-scale feature interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–11, 2024.
[45] X. Zhao, Z. Yang, Q. Li, and Q. Wang, "Parameter-efficient transfer learning for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–12, 2025.
[46] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, 2024.
[47] J. Luo et al., "When large vision-language model meets large remote sensing imagery: Coarse-to-fine text-guided token pruning," 2025, *arXiv:2503.07588*.
[48] Y. Hu, J. Yuan, C. Wen, X. Lu, Y. Liu, and X. Li, "RSGPT: A remote sensing vision language model and benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 224, pp. 272–286, Jun. 2025.
[49] Y. Zhan, Z. Xiong, and Y. Yuan, "SkyEyeGPT: Unifying remote sensing vision-language tasks via instruction tuning with large language model," *ISPRS J. Photogramm. Remote Sens.*, vol. 221, pp. 64–77, Mar. 2025.
[50] J. Luo et al., "SkySenseGPT: A fine-grained instruction tuning dataset and model for remote sensing vision-language understanding," 2024, *arXiv:2406.10100*.
[51] Y. Zhang et al., "SkySense v2: A unified foundation model for multimodal remote sensing," 2025, *arXiv:2507.13812*.
[52] J. Tang et al., "Remote sensing scene graph generation for improved retrieval based on spatial relationships," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 741–752, Feb. 2025.
[53] Y. Li et al., "STAR: A first-ever dataset and a large-scale benchmark for scene graph generation in large-size satellite imagery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 3, pp. 1832–1849, Mar. 2025.
[54] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and GeoRSCLIP: A large scale vision-language dataset and a large vision-language model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3449154.
[55] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.
[56] P. Anderson et al., "Bottom-up and top-down attention for image captioning and visual question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6077–6086.
[57] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 28, 2015, pp. 91–99.
[58] E. J. Hu et al., "LoRa: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022, vol. 1, no. 2, p. 3.
[59] N. Ravi et al., "SAM 2: Segment anything in images and videos," 2024, *arXiv:2408.00714*.
[60] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.
[61] P. Zhang et al., "InternLM-XComposer-2.5: A versatile large vision language model supporting long-contextual input and output," 2024, *arXiv:2407.03320*.



Xu Zhang received the B.S. degree from the Binjiang College, Nanjing University of Information Science and Technology, Nanjing, China, in 2015, and the M.S. degree from Nanjing University of Information Science and Technology in 2019. He is currently pursuing the Ph.D. degree with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China.

His research interests include computer vision and multimodal scene understanding.



Jiabin Fang received the B.S. degree from Southwest Petroleum University, Chengdu, China, in 2015, and the M.S. degree from the Stevens Institute of Technology, Hoboken, NJ, USA, in 2017. He is currently pursuing the Ph.D. degree with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China.

His research interests include autonomous driving and computer vision.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.



Zhuoming Ding received the B.S. degree from the College of Natural Resources and Environment, Northwest A&F University, Xi'an, China, in 2016, and the M.S. degree from the College of Computer Science and Informatics, Cardiff University, Cardiff, U.K., in 2023. He is currently pursuing the Ph.D. degree with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China.

His research interests include computer vision and vision–language large models.



Jin Yuan received the bachelor's and master's degrees from Fudan University, Shanghai, China, in 2005 and 2008, respectively, and the Ph.D. degree in computer science from the National University of Singapore, Singapore, in 2013.

He is currently an Associate Professor with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China. His research interests include multimedia retrieval and machine learning.



Xuan Liu (Member, IEEE) received the B.Sc. degree in information and computing mathematics from Xiangtan University, Xiangtan, China, in 2005, the M.Sc. degree in computer science from the National University of Defense Technology, Changsha, China, in 2008, and the Ph.D. degree from The Hong Kong Polytechnic University, Hong Kong, in 2015.

She is currently a Professor with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China. Her research interests include machine learning, the Internet of Things, and mobile computing.



Qianjun Zhang received the Ph.D. degree from Sichuan University, Chengdu, Sichuan, China, in 2017.

She is currently a Post-Doctoral Researcher at Southwest Jiaotong University Chengdu. Her research interests include artificial intelligence, multimodal processing, and large-scale models.



Zhiyong Li (Member, IEEE) received the M.Sc. degree in system engineering from the National University of Defense Technology (NUTD), Changsha, China, in 1996, and the Ph.D. degree in control theory and control engineering from Hunan University (HNU), Changsha, in 2004.

In 2004, he joined the College of Computer Science and Electronic Engineering, HNU, where he is currently a Full Professor. He has published more than 100 papers in international journals and conferences. His research interests include intelligent perception and autonomous moving body, machine learning and industrial big data, and intelligent optimization algorithms with applications.

Dr. Li is a member of China Computer Federation (CCF) and Chinese Association for Artificial Intelligence (CAAI).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:19:09 UTC from IEEE Xplore. Restrictions apply.